VOLTWYN VW–MTC1 MazeTrack Controller – Rev A

Objective wire–maze assessment controller for ataxia research: 4–ch Hall sensing, 12–bit ADC, IMU, touch detection, haptics, Li–ion power, USB–C, microSD, Wi–Fi/BLE.

Power

File: Power.kicad_sch

Power: USB–C, charger, power path, 3.3 V buck–boost

MCU

File: MCU.kicad_sch

MCU: ESP32–S3, UI, IMU, storage, expansion

Sensing

File: Sensing.kicad_sch

Sensing: Hall front end, 12–bit ADC, handle I/O, haptics

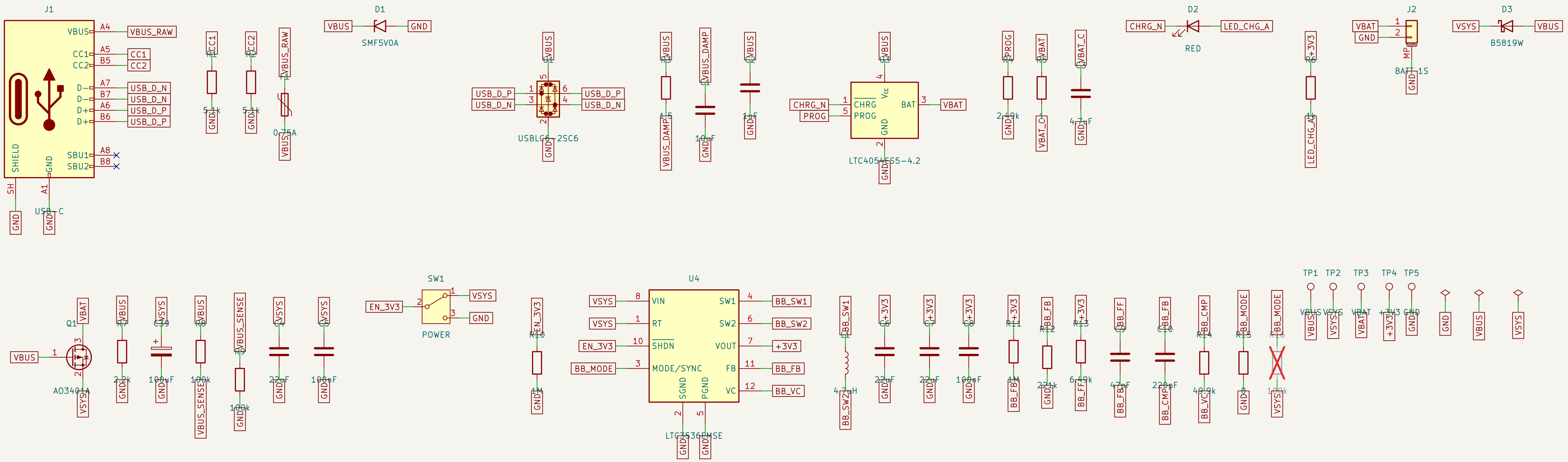
MazeIO

File: MazeIO.kicad_sch

Maze I/O: wire / start / finish touch detection, buzzer

Designed by S. Veeriah. Generated from design.py netlist spec
VW–MTC1 MazeTrack Controller – Root
Voltwyn
Sheet: /
File: VW–MTC1.kicad_sch
Title: VW–MTC1 MazeTrack Controller – Top level
Size: A3 Date: 2026–10–19 **Rev: A**
KiCad E.D.A. 10.0.6 Id: 1/5

Power: USB-C, charger, power path, 3.3 V buck-boost

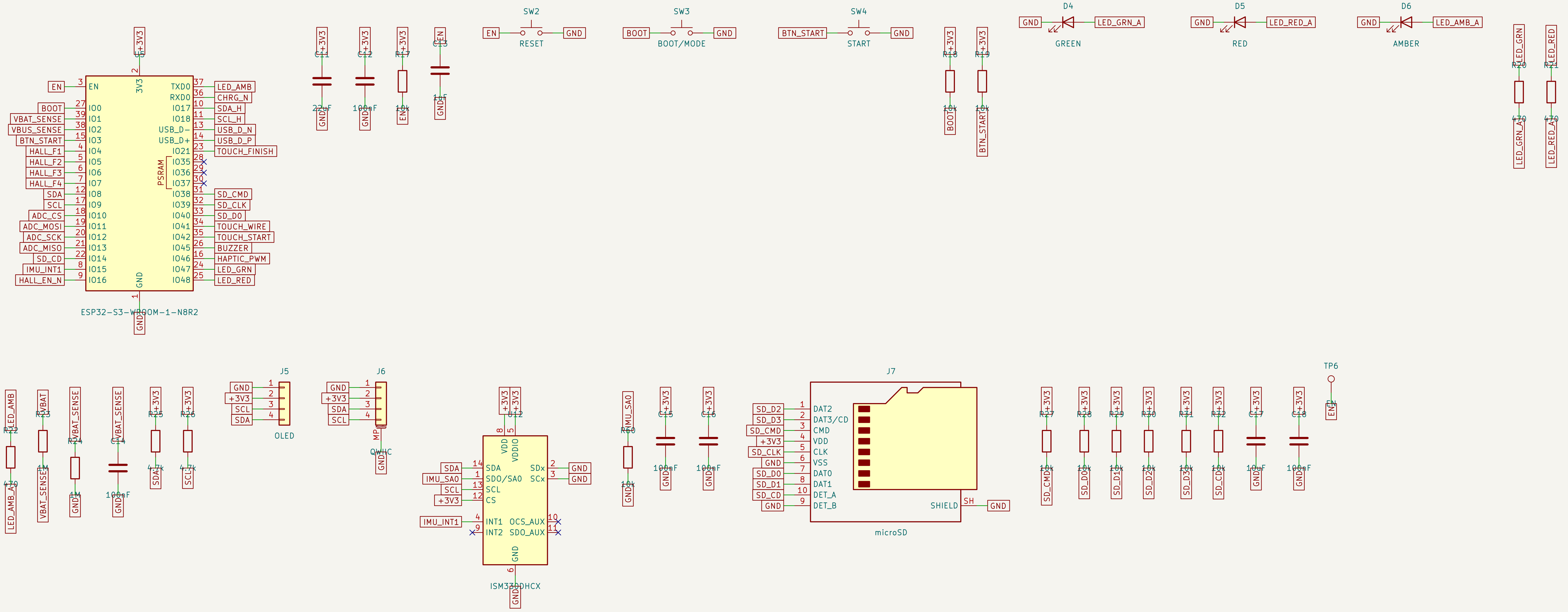


Designed by S. Veeriah. Generated from design.py netlist spec
VW-MTC1 MazeTrack Controller – Power
Voltwyn
Sheet: /Power/
File: Power.kicad_sch

Title: Power: USB-C, charger, power path, 3.3 V buck-boost

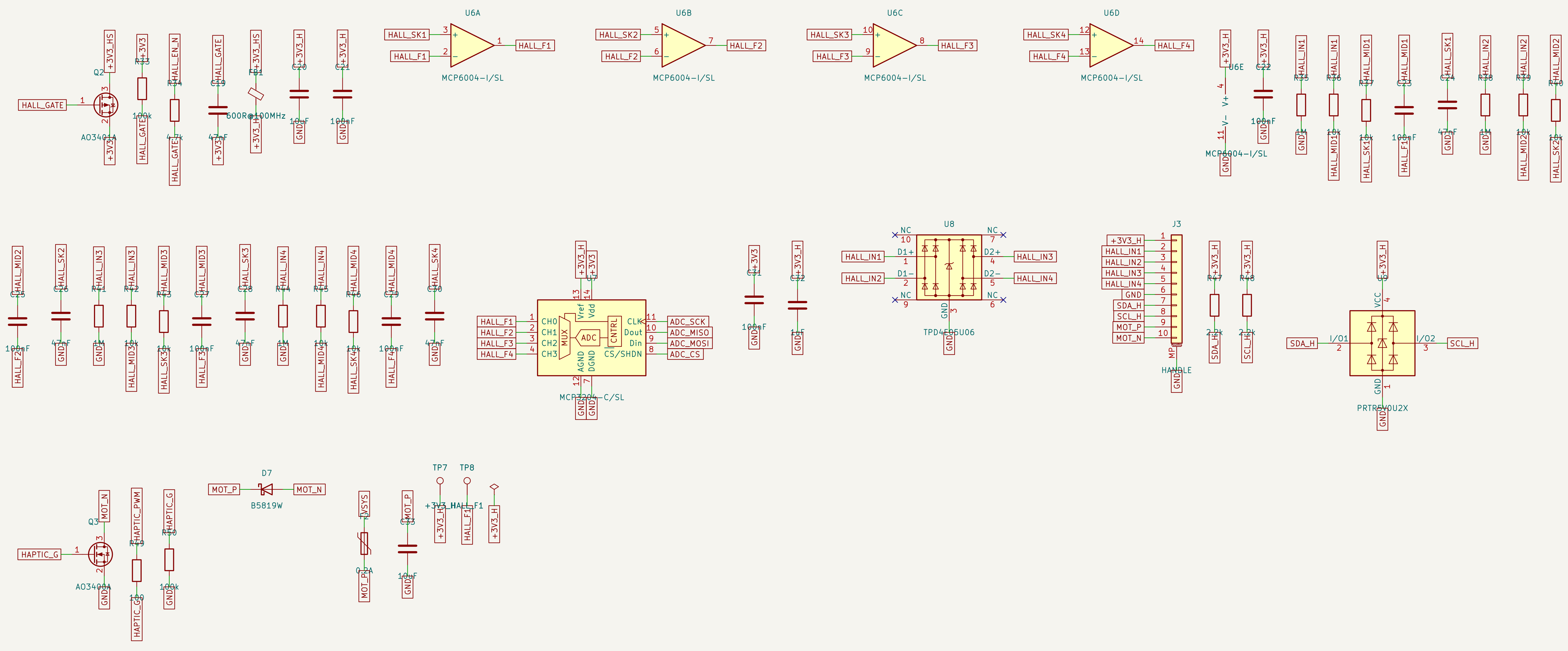
Size: A3 Date: 2026-10-19 **Rev: A**
KiCad E.D.A. 10.0.6 Id: 2/5

MCU: ESP32−S3, UI, IMU, storage, expansion



Designed by S. Veeriah. Generated from design.py netlist spec
VW−MTC1 MazeTrack Controller − MCU
Voltwyn
Sheet: /MCU/
File: MCU.kicad_sch
Title: MCU: ESP32−S3, UI, IMU, storage, expansion
Size: A3 Date: 2026-10-19 **Rev: A**
KiCad E.D.A. 10.0.6 Id: 3/5

Sensing: Hall front end, 12-bit ADC, handle I/O, haptics



Designed by S. Veeriah. Generated from design.py netlist spec
VW-MTC1 MazeTrack Controller – Sensing

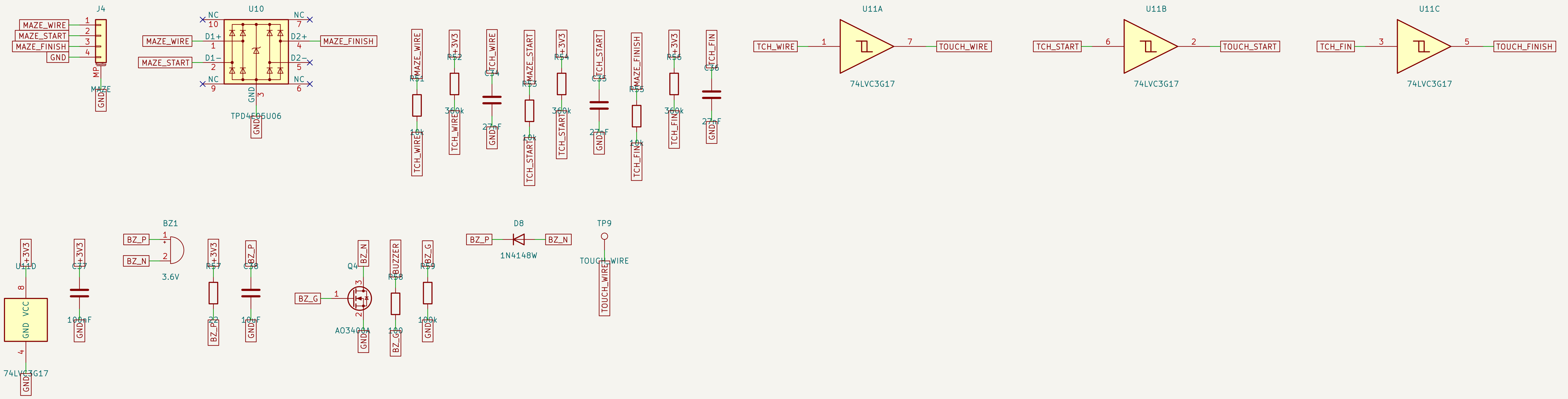
Voltwyn

Sheet: /Sensing/
File: Sensing.kicad_sch

Title: Sensing: Hall front end, 12-bit ADC, handle I/O, haptics

Size: A3 Date: 2026-10-19 **Rev: A**
KiCad E.D.A. 10.0.6 Id: 4/5

Maze I/O: wire / start / finish touch detection, buzzer



Designed by S. Veeriah. Generated from design.py netlist spec
VW−MTC1 MazeTrack Controller − MazeIO
Voltwyn
Sheet: /MazeIO/
File: MazeIO.kicad_sch

Title: Maze I/O: wire / start / finish touch detection, buzzer

Size: A3	Date: 2026−10−19	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/5